

Remote Sensing Data Mining for Monitoring Frozen Peatland Ecosystems in the Hudson Bay

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1 Context

Peatlands store more than a third of the global soil organic carbon stock. The Hudson Bay Lowlands, located in northern Canada, represent one of the largest continuous expanses of peatlands between 60° and 70°N. In recent decades, this area has faced strong climate warming : air temperatures have increased at more than twice the global average, contributing to the rapid degradation of discontinuous permafrost.

In Wapusk National Park, peat plateau bogs form the dominant ecosystem, with variations in vegetation cover and surface moisture closely linked to underlying patterns of discontinuous permafrost, shaping a mosaic of distinct ecological zones. As thaw progresses, peat plateaus collapse, giving way to wetter environments colonized by Sphagnum mosses. These transitions result in major changes in carbon dynamics, methane fluxes, vegetation composition and hydrology.

Remote sensing provides essential tools for monitoring these rapidly changing and spatially fragmented ecosystems. However, the fine-scale mosaic of dry, lichen-covered plateaus and waterlogged Sphagnum habitats presents challenges for conventional satellite-based classification. To address this, our study leverages ultra-high-resolution drone imagery alongside super-resolved Sentinel-2 time series data. This multi-scale approach enables the detailed detection and mapping of subtle vegetation and hydrological changes across the landscape.

2 Objectives

This study aims to advance the monitoring and classification of frozen peatland ecosystems in Hudson Bay by integrating multi-scale remote sensing data, including high-resolution drone imagery and super-resolved satellite observations. The primary study area is the degraded peat plateau of Twin Lakes, near the Churchill research camp, where RGB, thermal, and digital surface model drone images will be used and compared with Sentinel-2 data. The main objectives are to classify peatland ecozones at high resolution and to evaluate water surfaces and vegetation transitions in relation to permafrost thaw over the large areas covered by Sentinel-2 images. The methodology will then be extended to additional drone and satellite surveys in other study areas within Wapusk National Park. This work contributes to improving remote sensing approaches for Arctic ecosystem

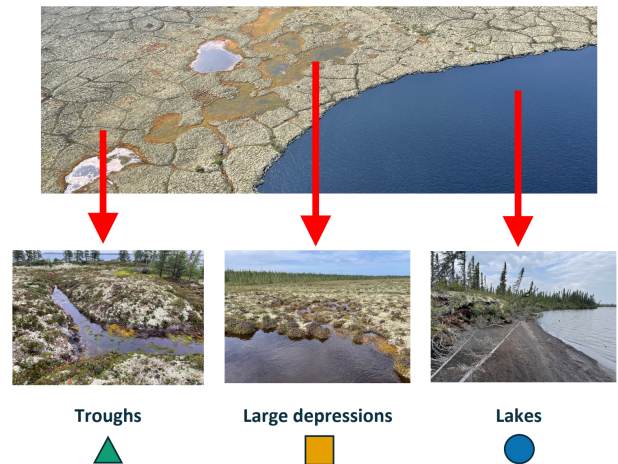


Figure 1: Peat plateau fragmented ecosystems in Wapusk National Park

monitoring and provides valuable data to support ongoing studies on ecological transformations in peatlands.